Bulgarian Diploma Thesis

Used Cars Price Predictor

with Machine Learning

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Declaration for Senior Project and Diploma Thesis

Title: Used Cars Price Prediction

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Car valuation has usually relies on manual analysis of the market or static pricing models. These methods often produce inconsistent or outdated estimates, because they fail to find the connection between the different vehicles and their features. The goal of this project is to address these limitations by developing a full-stack web application. The application is going to be using the following machine learning algorithms for predicting the prices of used cars - Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting. The core of the project is the backend and it is implemented in ASP.NET Core. The frontend is developed with React and TypeScript. The backend handles model training, evaluation, and serialization. The frontend provides an intuitive interface for predictions, trend visualization across time ranges, and side-by-side model comparisons via charts. In the data preprocessing pipeline there are data quality checks that verify for missing values and outliers. Also there is feature encoding and normalization. The models are comapred and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2). Ridge regression combined with ensemble residual learners (Random Forest and Gradient Boosting) gave the most accurate performance.

Declaration of authorship:

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